

### ■ 2.5.6 Functions that Return Multiple Values

It is sometimes convenient to write a function that returns several values at the same time. You can do this in *Mathematica* simply by making the function return a list of expressions. A convenient way to pick out each of the expressions in this returned list is just to assign the list as the value of a list of variables.

This assigns a the value 1 and b the value 4.

```
In[1]:= {a, b} = {1, 4}
Out[1]= {1, 4}
```

Here is a function that returns a list of three values.

```
In[2]:= f[x_] := {x, 1-x, 1+x}
```

This assigns the variables a, b and c to be the three values returned by f.

```
In[3]:= {a, b, c} = f[5]
Out[3]= {5, -4, 6}
```